Brazilian Amazon Forest Regrowth Rates Summer Research Project Paper and Summaries

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Introduction

The Brazilian Amazon is one of the ecologically richest places on Earth, containing the largest tropical rainforest in the world and supplying vital ecosystem services that extend far beyond South America. The forest holds vast amounts of carbon, regulates regional and global climate through moisture recycling, maintains an incomparable diversity of life, and supports the livelihoods and cultures of numerous Indigenous peoples. Because of these ecosystem services, changes in land cover in Amazonia have consequences that extend beyond Brazil, affecting global climate regulation, carbon cycling, freshwater provision, and species conservation. Despite its value, the Amazon has suffered decades of deforestation and land cover conversion due to agriculture, cattle ranching, logging, infrastructure development, and mineral extraction. While Brazil has shown that strict environmental regulation and enforcement could drastically reduce deforestation in the early 2000s, recent political and economic changes have made enforcement efforts weaker in many regions of the Amazon, allowing the rate of deforestation to increase again. Researchers and policymakers have pointed out that ongoing forest conversion could push certain parts of the Amazon towards ecological tipping points from which recovery would be increasingly difficult.

Mining is among several factors leading to land cover change; it removes vegetation, changes soils, fragment habitats, and affects water cycle. Contrary to many agricultural land uses, mining usually leaves behind a degraded landscape that takes many decades of natural succession or active restoration to recover ecological functions. The expansion of pasturelands has also become a key factor shaping the Brazilian Amazon, as significant expanses of virgin forest have been converted into pastures to allow grazing cattle. While both mining and pasture lead to forest destruction, their ecological impacts differ vastly. Therefore, it is necessary to know how fast

forests can recover from different types of anthropogenic disturbances in order to estimate ecosystem resilience, carbon stock, priorities in restoration efforts, and conservation policy. Recent changes in Brazilian environmental policies highlight the need to investigate forest recovery processes. Under President Jair Bolsonaro, decreased environmental enforcement, weakened protection of Indigenous territories, and greater support for resource extraction were accompanied by increased deforestation rates and pressure to expand mining operations within the Amazon region. NGOs and researchers have registered the decrease in enforcement activities and increase in land clearing activities, while the proposed expansion of mining activities within Indigenous lands raised concern regarding ecological consequences and the destruction of intact forest ecosystems. These developments make it crucial to investigate not only how forests are degraded but also whether they can return to their original state.

For these purposes, this paper uses annual land cover classifications for 1985 through 2024 from the MapBiomas Collection 10 dataset. MapBiomas provides one of the best databases of land use and land cover change for Brazil, applying Landsat satellite images together with machine-learning classifications techniques to obtain annual maps of land cover changes in Brazil. Since the database gives land cover classifications for each year, it is possible to follow individual pixels throughout the whole period and see when they first become anthropogenic land uses, if and when they recover to natural state, and calculate recovery periods. The annual basis of the dataset allows us to trace the dynamics of recovery and compare the dynamics of recovery of different pixels. MapBiomas is becoming a vital tool for monitoring deforestation, validating land cover changes, and supporting environmental enforcement in Brazil.

Using these annual land cover data, this research investigates the recovery of forest pixels after anthropogenic disturbance, particularly after mining and pasture. Data analysis was performed in R programming language, and it allowed us to calculate recovery periods, identify forest pixels that turned into anthropogenic land uses, then returned to Forest Formation, and visualize these transitions in histograms, annual recovery plots, density plots, and alluvial diagrams. Statistical analysis, including analysis of variance (ANOVA) and chi-squared tests, was used to investigate whether recovery periods of pixels in mining and pasture disturbed landscapes differed. Instead of using summary variables indicating the first anthropogenic land use, this paper traces annual

land cover changes of pixels throughout the entire 40-year period to detect all mining and pasture disturbance events.

The main hypothesis of this research is that recovery dynamics after forest destruction differ according to the kind of land use that caused the disturbance; that is, mining results in slower forest recovery and a lower return rate to Forest Formation than pasture. By quantifying these dynamics for almost four decades of satellite images, this paper aims to contribute to the knowledge about forest resilience after anthropogenic disturbances and illustrate how long-term remote sensing datasets can be applied to assess restoration success. The findings of this work can be useful for discussion of sustainable land management, ecological restoration, biodiversity conservation, and environmental policy in the Brazilian Amazon. As mining, agriculture, and infrastructure development continue to transform tropical forests, it is critical to understand their capacity for natural recovery in order to conserve one of the most valuable ecosystems on the planet.

Methods

We used annual land use and land cover transition for a test region on the eastern edge of Munduruku territory (7.75°–7.25° S, 56.95°–56.45° W) from MapBiomas Brazil Collection 10.1 available in Google Earth Engine. The dataset provides annual 30-m classifications from 1985 through 2024. Anthropogenic land uses include plantation forestry, pasture, agriculture and its subclasses, mosaics of uses, urban areas, other non-vegetated areas, mining, and aquaculture. We limited the analysis to pixels whose land cover class before anthropogenic conversion was natural vegetation, including forest, savanna, mangrove, floodable forest, wetland, grassland, and wooded or herbaceous sandbank vegetation.

For each pixel, we identified the year when it first changed from natural vegetation to anthropogenic use, recorded the previous land cover class, the initial anthropogenic land cover class, and the year of conversion. Pixels already classified as anthropogenic in 1985 were excluded from analysis because it was impossible to identify their previous land cover class. Then we found the first year when the pixel returned to its exact pre-conversion land cover class. The resulting sample consisted only of pixels that underwent anthropogenic conversion and

returned back. For each pixel in the sample, we calculated the number of years between conversion and return and extracted pixel coordinates, land cover transition attributes, and the entire 1985–2024 land cover history for further analysis in R. We also summarized the area of return in hectares for each original land cover class, initial anthropogenic land cover class, and return year.

After the processing in Google Earth Engine was finished, the dataset was processed in R in order to analyze both broad land cover change and recovery dynamics of individual pixels. Every pixel in the dataset included annual land cover classifications from 1985 through 2024, and this allowed us to investigate the whole 40-year period instead of analyzing only the first and last years. The first set of plots was created to provide information about the general land cover composition of the study area. Pie charts and bar charts were plotted for 1985, 1995, 2005, 2015, and 2024 to compare the proportions of different land cover classes at approximately ten-year intervals. These plots helped to understand whether the composition of the landscape changed through time and whether anthropogenic classes like mining, pasture, agriculture, or other developed land uses gained or lost their representation in the landscape. Bar charts were especially useful because every MapBiomas pixel is 30 m × 30 m in size, so the number of pixels in each land cover class could be considered the area of that class.

Since mining was one of the main land uses under investigation, the second set of plots was devoted to the temporal dynamics of mining. Mining pixel count was calculated from the entire set of annual land cover columns and plotted as a time series. This plot helped to find periods of rapid growth or decline of mining and whether they were associated with some political or environmental period. Another analysis was aimed to identify pixels classified as Forest Formation in 1985 and then find out how many of them became Mining pixels annually thereafter. The purpose of this separation was to isolate the general dynamics of expansion of mining from the dynamics of forest conversion to mining. With the help of these two plots, we could find out whether the growth of mining was mainly the result of expansion into the forested areas or the consequence of changes within the area already occupied by another land use. Recovery analyses were conducted in order to find out whether the loss of forests as a result of anthropogenic disturbances is temporary or persistent. The recovery period was calculated as the

difference between the year of first entrance to the anthropogenic class and the year of return to the previous land cover class. Histograms were used to visualize the distribution of recovery periods because they showed not only the most frequent recovery periods but also long periods of recovery. The first mining-recovery analysis included pixels whose land cover class before the anthropogenic class was Forest Formation, whose first anthropogenic disturbance was Mining, and that returned to Forest Formation. A stricter version of this analysis involved only pixels classified as Forest Formation in 2024. Comparison of these two sets of pixels helped to distinguish between pixels that temporarily returned to the forest state and pixels that maintained their forest classification through the entire observation period. The calculation of means, medians, standard deviations, and other summary statistics was also performed to find the characteristics of recovery periods and unusual recovery periods.

We created annual and cumulative recovery plots to examine not the number of years but the timing of forest return. The number of pixels disturbed by mining that returned to Forest Formation was calculated each year, and cumulative values were calculated up to 2024. These plots helped to find out whether the forest recovery happened consistently through time or occurred during specific periods. The difference is important because two disturbances could have the same mean recovery period but completely different recovery periods dynamics. Cumulative recovery curves also allowed us to assess the proportion of recovery of disturbed landscape in different periods.

To put mining into a broader perspective, recovery periods were also compared for other anthropogenic land use classes. Histograms and density plots were created based on recovery periods of pixels that were disturbed by mining, pasture, agriculture, forest plantation, mosaic land use, urbanization, and other major anthropogenic land uses. Density plots were used along with histograms because they allowed to compare the shapes of multiple recovery period distributions on one scale. The aim of this analysis was to test whether mining produces a unique recovery pattern compared to other land use conversions. Faceted histograms were created to analyze recovery periods for pixels with different anthropogenic disturbances in order to avoid influence of sample size and distribution shape by one dominant anthropogenic class.

The annual land cover columns were also transformed from wide to long format to examine the entire land cover history of each pixel through time. An alluvial plot was used to visualize the transitions between anthropogenic land cover classes between 1985 and 2024 while grouping non-anthropogenic land cover classes into one natural category. This visualization helped to find out whether pixels followed simple transitions, such as Forest Formation to Mining and back to Forest Formation, or more complex transitions through multiple anthropogenic land covers. This step was important because summary variables like the first anthropogenic land cover class do not capture all transitions that happen. A pixel could, for instance, transition from Forest Formation to Pasture, then become Mining, and finally return to Forest Formation. Thus, tracing the entire annual land cover history reduced the possibility of misattributing the recovery to the first disturbance.

Pasture was chosen as a comparative anthropogenic land use for mining because they both convert Forest Formation but represent very different disturbances. Using the entire annual land cover history, pixels that were disturbed by Mining but not Pasture and pixels that were disturbed by Pasture but not Mining were split into two distinct groups. For each group, the year of first disturbance, the year of first return to Forest Formation, and the recovery period were calculated. These recovery periods were compared visually using violin and box plots and statistically using analysis of variance (ANOVA) or Welch's ANOVA in case of unequal variances and two-sample tests of mean recovery period. This analysis was aimed to check whether the pixels disturbed by mining required more time to return to Forest Formation than those disturbed by pasture. The effect sizes were calculated in addition to statistical significance to assess the magnitude of this difference.

Chi-squared test was used to test another but different question – whether the probability to return to Forest Formation by 2024 was higher for mining- or pasture-disturbed pixels. Pixels were categorized as recovered or not recovered by the end of the dataset period, and recovery proportions were compared between the two disturbance types. In addition, 1-5 years, 6-10 years, 11-20 years, and more than 20 years categories were analyzed to compare the recovery speed distribution between disturbances. Thus, statistical and visual analyses were used to test two aspects of ecosystem resilience – the probability to recover and the recovery time.

Conclusions

This analysis examines land cover change between 1985 and 2024, focusing on the expansion of mining activity, the fate of forest converted to mining, and how fast disturbed pixels return to forest cover. Land cover was analyzed at roughly ten-year intervals (1985, 1995, 2005, 2015, 2024), and pixel-level transition history was analyzed to model the timing and duration of forest loss and recovery. The analysis reveals two storylines – the pasture expansion in the middle of the period and more recent rapidly accelerating mining expansion.

Land Cover Change, 1985–2024

In 1985, the landscape was predominantly forested, with Forest Formation taking the largest part of pixels, while only small portions were covered by grassland, savanna, floodable forest, and wetland. By 1995, pasture became a considerable land cover class, and by 2005 it became the largest class, covering more than half of the total area, while forest cover shrank. This trend was reversed between 2005 and 2015, with Forest Formation cover bouncing back to levels roughly equal to those of Pasture, indicating either reduced deforestation pressures, land abandonment or regrowth of former clearance sites. By 2024, Forest Formation cover regained its status as the predominant cover class, but this seemingly positive turn conceals a major compositional change: mining operations, negligible through the mid-2000s as a fraction of the landscape, had become a prominent feature of the 2024 land cover composition.

The Recent Acceleration of Mining

The trend of mining pixels through 1985 to 2024 shows a mild growth until mid-1990s, then stabilization and a gradual decrease during the 2000s, followed by an abrupt and steady increase that started in the mid-2010s. Pixel counts of mining in the last years of the series are several times greater than in any prior years of the forty-year period, and the trend of increase appears to be accelerating rather than slowing down. Mining pixel counts obtained from direct conversion of Forest Formation pixels exhibit a parallel trend, with a mid-1990s increase, followed by a low 2000s, and then an abrupt increase after roughly 2015, reaching its maximum values only recently. This suggests that the current period is not a continuation of historical mining activity, but a new, intensifying phase of forest-to-mining conversion. Given the timing, this acceleration may be attributable to the regulatory and enforcement changes mentioned in the research

background context. This trend is the most policy-relevant finding in the dataset, and represents a major change relative to the historical baseline: mining deforestation is ongoing and worsening.

Recovery Dynamics of Disturbed Land

The recovery dynamics of disturbed pixels that were converted to mining and subsequently returned to forest is relatively quick, with right-skew: the median recovery time is approximately five years, the mean is pulled upward slightly, to around six to seven years, due to a small percentage of pixels with extremely long recovery times, up to 31 years. The peak recovery time falls between three and five years, and the number of pixels recovered from mining by each year shows two spikes, in the late 1990s and around 2019 to 2020, followed by a precipitous decline in the most recent years. The most reasonable interpretation of the latter trend is the right censoring of recently disturbed pixels: they simply did not have enough time to recover and thus are not counted in the recovery time series. It should therefore be understood that the cumulative graph starts to flatten out in 2020 because it takes many years for mining to recover, and the most recent mining pixels simply have not had enough time yet. When the recovery time is considered among all disturbed pixels, not just mining, the picture changes: the median recovery time increases to about 13 years, with the mean of around 11.6 years, the standard deviation of approximately 6.6 years and maximum of 36 years, almost twice the median recovery time for mining pixels. Thus, despite the fact that mining pixels are the most rapidly growing cause of deforestation in recent years, it is not the longest-lasting land cover class.

Recovery Time by Anthropogenic Land-Use Class

Comparison of the distributions of recovery times between anthropogenic classes explains the difference. Of the 64,804 disturbed pixels with identified first anthropogenic use, the vast majority of them (55,226 pixels) were used for pasture, mining made up a substantial minority (9,533 pixels) and Other Non-Vegetated Areas contributed a negligibly small share (45 pixels). The pasture pixels are distributed bimodally, showing one cluster recovering fast (within five years) and another one taking about 11 to 20 years; 29,659 out of 51,796 pasture-recovery events fall into the latter group. Mining pixels, on the contrary, are mostly in the 1-to-5 year band (4,621 out of 8,025 pixels), while the share of those with recovery time over 10 years is much smaller. A chi-squared test shows that recovery-time category and land-use class are significantly

associated ($\chi^2 = 6985.1$, df = 3, $p < 2.2 \times 10^{-16}$): mining pixels are approximately 2.3 times as frequently represented in the fastest recovery time band, and pasture pixels – vice versa. Effect size is very small (Cramer's V ≈ 0.04), due to the huge sample size needed to reach statistical significance, but the pattern is clear and consistent: mining pixels recover quicker and pasture pixels recover slower, although the latter make up the majority of disturbed pixels. A boxplot/violin plot comparing the two groups' recovery times confirms this: the distribution of mining recovery times is shorter with the median close to 4 years and right-skewed, while that of pasture is higher and wider, going well beyond 30 years.

Nearly Complete Recovery, With a Small but Meaningful Gap

One of the most encouraging findings of the dataset is the fact that nearly all disturbed pixels eventually return to forest by the end of the period studied. 99.99% of pasture pixels (51,790 out of 51,796) had recovered to Forest Formation cover by 2024; 99.70% (8,001 out of 8,025) of mining pixels had recovered. While in absolute terms unrecovered pixels are rare in both categories, mining pixels are significantly more likely not to recover: the odds ratio analysis puts odds of non-recovery for mining about 26 times greater than for pasture (odds ratio ≈ 25.9). This is explained by the fact that mining is the more recent disturbance, thus the greater proportion of them are still within the observation window, while the older pasture disturbance has allowed nearly all of its disturbed pixels to recover regardless of the recovery time.

Political and Regulatory Context for the Post-2015 Acceleration

Timing of the acceleration of mining fits perfectly into a well-documented sequence of political and regulatory change in Brazil. During President Dilma Rousseff's term (January 2011-August 2016), economic development became a priority over environmental protection and rights of Indigenous peoples; large infrastructure and energy projects were favored by the government. After Rousseff's impeachment, Michel Temer's administration heavily relied on the ruralista Congressional bloc representing agribusiness and resource-development interests. This government is associated with decreased conservation safeguards, shrinking of protected areas and slowed process of recognition of Indigenous territories. 2018's election of Jair Bolsonaro marked a new step of the process: his administration openly criticized the existing environmental and Indigenous regulations, placed friendly towards extractive development officials in key

positions and advocated opening the Indigenous lands to mining. The sequence of gradually weakening protection from 2011-2015, additional pressure from 2016-2018 and rapid acceleration from 2019 onwards matches perfectly with the inflection point in the mining pixel count around 2015 and subsequent increase. While this analysis cannot prove causality, the fit between the political timeline and the land-cover timeline is an interesting and policy-relevant pattern.

Composition of Annual Disturbance and Censoring Patterns

Decomposition of the pixel transitions by year (Mining, Pasture, and Natural/Other) shows that the trend described above: while pasture's share of annually disturbed pixels grew until the mid-1990s, reached its peak in the mid-2000s and decreased afterwards, mining's share, relatively small and stable through most of the time period, expanded noticeably in the final years shown. Annual count of pasture pixels recovering to forest also exhibits a similar trend, having a growth until mid-1990s, a peak in the mid-2010s (approximately 6,000 pixels recovered in the biggest year, around 2017-2018) and then declining to near-zero by 2024. The reason for the decline is likely the decreasing number of unrecovered pixels of the older pasture disturbance, coupled with the same censoring of recently disturbed pixels seen in the mining data, rather than a reduction in the actual regrowth rate, given the complete recovery rates for pasture (99.99%) and that pasture disturbance is, on average, older than mining.

Limitations

The precipitous decline in recovery rates for both mining and pasture after around 2020 is very likely a right-censoring artifact of the observation period (data up to 2024). Classification of the land cover by year at the pixel level also involves measurement error, particularly in the case of distinguishing early regrowth from other vegetation types. Chi-squared and odds ratio results are merely association tests, no causation can be established. The very large sample size causes even small practical effects (Cramer's $V \approx 0.04$) to become highly statistically significant. Finally, the fit between the post-2015 mining acceleration and the political timeline in Brazil is a suggestive, observational one: it has not been tested against alternative explanations such as commodity price fluctuations or infrastructural development, and the class of Other Non-Vegetated Areas is too small (45 pixels) to allow a reliable

Conclusion

The land cover dynamics 1985 to 2024 reveal a two-step pattern: pasture-driven deforestation and nearly complete restoration from the early 1980s to roughly 2010 followed by rapid post-2015 mining development that appears to outstrip previous transitions in speed and parallels well-documented weakening of Brazilian environmental and Indigenous land protections during the same period. Luckily, both pasture and mining eventually restore the majority of disturbed forest areas, reaching 99.5% restoration rates for both types of disturbance by 2024. However, pasture restoration is a slow process on a per-pixel level but represents a diminishing share of new disturbance while mining is fast on a per-pixel level but disturbances happen on an increasing scale; the future trend depends on whether the rate of recovery can keep up with mining acceleration. Additional observation past 2024 is required to determine whether the apparent 2020-2024 stagnation is an anomaly or just a delayed reaction of recently disturbed pixels.

100-Word Summary:

This study used annually-released MapBiomas Brasil Collection 10.1 to track the individual forest pixels in the Brazilian Amazon Munduruku region, analyzing how many years of conversion to mining or pasture it takes to recover from these activities in the region. While pasture historically represented most of deforestation in the region and is a slow activity to recover from, the recent acceleration of mining in the region has resulted in much quicker recovery (median of 5 vs 13 years, respectively) and a high percentage of unrecovered forest. However, mining, despite being much quicker to recover, disturbs the land at an increasing rate while pasture has been recovering from that.

2-Page Summary:

Background

The Brazilian Amazon contains immense carbon stores and regulates global climate, while its biodiversity ensures unique way of life for Indigenous communities that live there; hence, land use changes in this region are highly impactful worldwide. Agricultural activities including ranching, logging, and mining have been actively driving deforestation and land degradation in the region in the recent decades; though, the enforcement of strong policies in the early 2000s helped to stabilize this situation for some time. Nonetheless, the recent political changes have brought mining back on the rise and, according to the data, forested land cover in the region suffers as a result. The aim of this study is to compare the impact of mining to pasture as disturbance on forest recovery rates; in particular, it is hypothesized that the former one is associated with significantly longer recovery periods than the latter.

Data and Methods

This study relies on publicly available annual 30-m resolution land cover dataset, MapBiomas Brasil Collection 10.1 1985 – 2024; the analysis area is a small portion of the Munduruku

territory in the Brazilian Amazon located to the east of the river. Because it gives classifications on annual rather than biannual basis, the data set allows to monitor the pixels' states across the years in the whole period. To assess land recovery period for mining- and pasture-disturbed pixels, the analysis determined for every pixel the year when it changed from the natural vegetation to an anthropogenic cover, and identified the moment it recovered; the latter is understood as changing back to pre-mining/pasture class. Data processing took place in Google Earth Engine and R environment, while data visualization included land cover pie and bar charts at about ten-year intervals, cumulative and yearly plots of mining and pastures, recovery-time distribution plots and alluvial diagrams of land cover change; also the box-violin plot for the comparison was constructed, while statistical tests for the comparison included ANOVA (plus Welch's ANOVA when required) for the mean recovery period and chi-square test for the distribution and recovery rate difference between mining and pasture.

Land Cover Changes 1985-2024

Initially in 1985, the region was mostly forest-covered. Then pasture started increasing in the area to become the dominant cover by 2005, occupying more than half of all pixels, as the forest shrinks. After 2005, the forest partly recovered by 2015, reaching roughly equal coverage with pasture; in 2024 the largest proportion of the land is covered by forest again. Nevertheless, it masks an important shift in land use: mining, which has been insignificant before mid-2000s and barely noticeable until 2015, becomes prominent enough in 2024 to become clearly noticeable land use in the region. In the words, the landscape undergoes the transformation from pasture-dominated to mining-influenced one.

Recent Acceleration of Mining in the Region

The number of mining pixels increases through the mid-1990s and plateaus and slightly decreases in the following decade until 2010. After that it rapidly increases until 2024 at least doubling previous peak values in the dataset. Forest pixels that convert directly to mining also show the same pattern, indicating that it marks a beginning of a distinct and intensive phase of

mining. These changes in mining match well a documented reduction of environmental and Indigenous lands protection during the past decade or so. In particular, during the presidency of Dilma Rousseff (2011-2016) the policy of environmental enforcement was significantly relaxed; during the time of Michel Temer (2016-2018), who represented the ruralista bloc interests, institutional pressure on the protection increased. Finally, the explicit policy for opening Indigenous and protected lands to mining and exploitation is implemented in the current administration of Jair Bolsonaro; hence, these observations show an interesting parallelism, though this study cannot make a direct causal inference from them.

Recovery Rates and Their Distribution

For mining-converted pixels recovering to forest the recovery period is fast with a right-skew: median approximately 5 years and mean about 6-7 years, with a maximum near 31 years. However, when it comes to combined disturbance pixels the median of the recovery period rises to 13 years (the mean is about 11.6 years, standard deviation 6.6 and the maximum – 36 years); that makes twice faster as for the mining pixels; hence, it can be said that mining despite being the source of the latest disturbances is not the slowest class on this parameter. Comparison helps to reveal what drives this difference. In particular, mining is heavily centered in the range of 1-5 years, while pasture shows bimodal recovery times; specifically, most of pasture pixels (51796 out of 90859 in total, 57%) recover in 11-20 years. It turns out that chi-square test (x^2=6985.1, df=3, p<2.2×10^-6) confirms that the association is significant; nonetheless, due to large sample size, Cramer's V is low – it is only 0.04. Recovery rate for both types of disturbance is almost complete – 99.99% and 99.70% of pasture and mining pixels, respectively. Also mining is still about 26 times less likely to be unrecovered (odds ratio = 25.9) because it represents the new and still active source of disturbance; hence, its last pixels do not have enough time to recover yet. Apparently the decreased number of recovery records for both classes after the 2020 period can probably be attributed to the fact that some pixels just lack enough time.

Conclusion and Implications

The 40-years period of land cover observation revealed a two-stage story: pasture-driven deforestation and nearly complete recovery from 1985 to roughly 2010 followed by the rapid acceleration in mining since 2015. Mining and pasture are sources of different disturbances: former one disturbs faster on a pixel level but the latter one represents a smaller share of it currently. The future direction of changes in the region will depend on whether recovery can catch up with mining disturbance. Limitations of the study include observational rather than causal nature of political discussion, measurement uncertainty of pixel-level classification, small sample of Other Non-Vegetated Areas and right-censored observation for the most recent disturbance cohort.